

$$G_o(s) = \frac{3}{s+4} = \frac{k}{sT+1}$$

$$k=? \quad T=? \quad y(t)=?$$

$$r(t) = 1_r(t)$$

$$k = \frac{3}{4}; \quad T = \frac{1}{4}$$

$$G_o(s) = \frac{3}{4(\frac{1}{4}s+1)} = \frac{\frac{3}{4}}{\frac{1}{4}s+1}$$

$$y(t) = k(1 - e^{-\frac{1}{T}t}), t \geq 0 \quad y(t) = \mathcal{L}^{-1}\{G_o(s)R(s)\}$$

$$= \frac{3}{4}(1 - e^{-4t}), t \geq 0$$

$$G_o(s) = \frac{G(s)}{1+G(s)} = \frac{3}{s+4} \Rightarrow G(s) = \frac{3}{s+1}$$

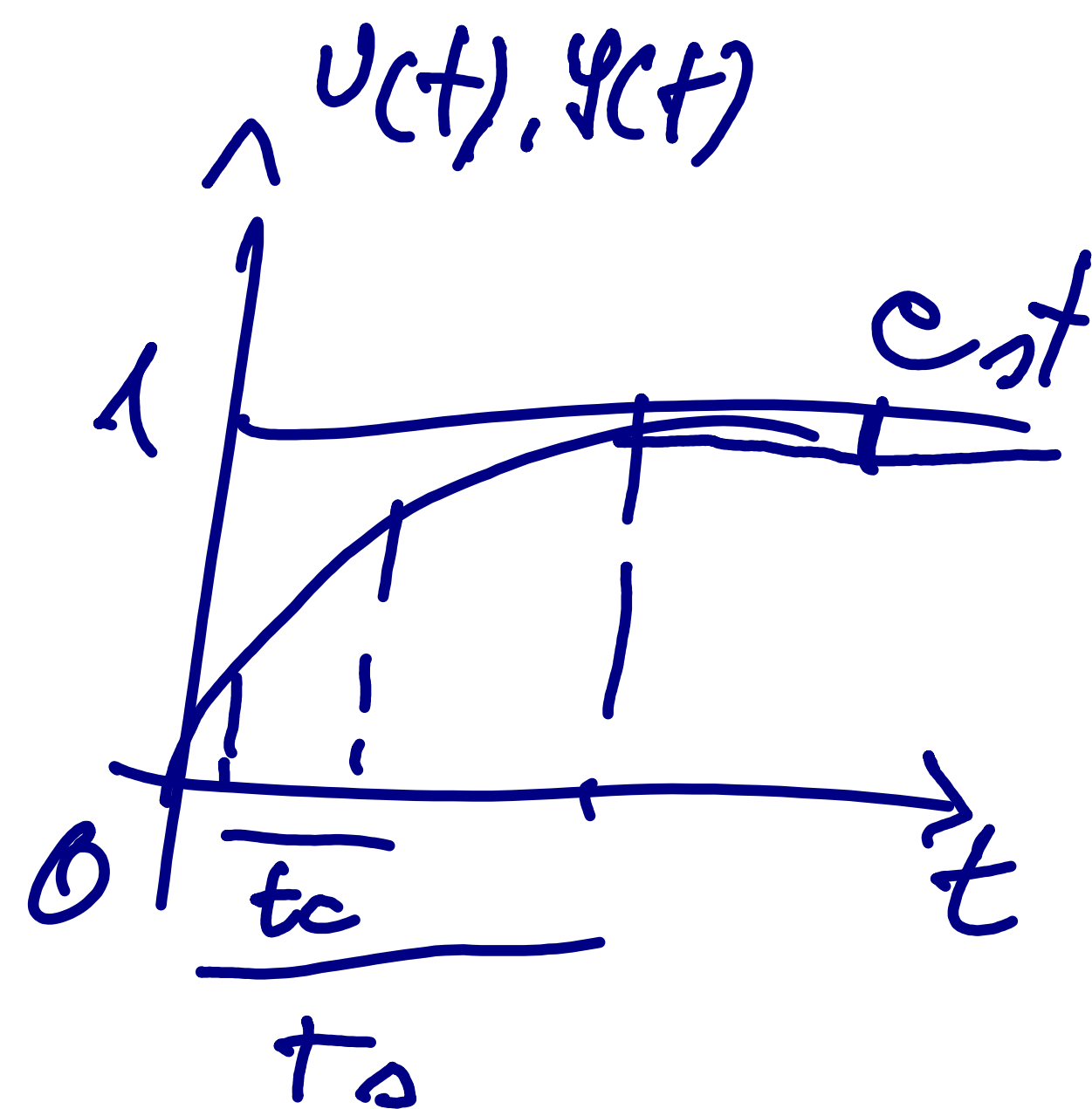
$$e_{st} = \frac{1}{1+k_p} = \frac{1}{4} = 0.25$$

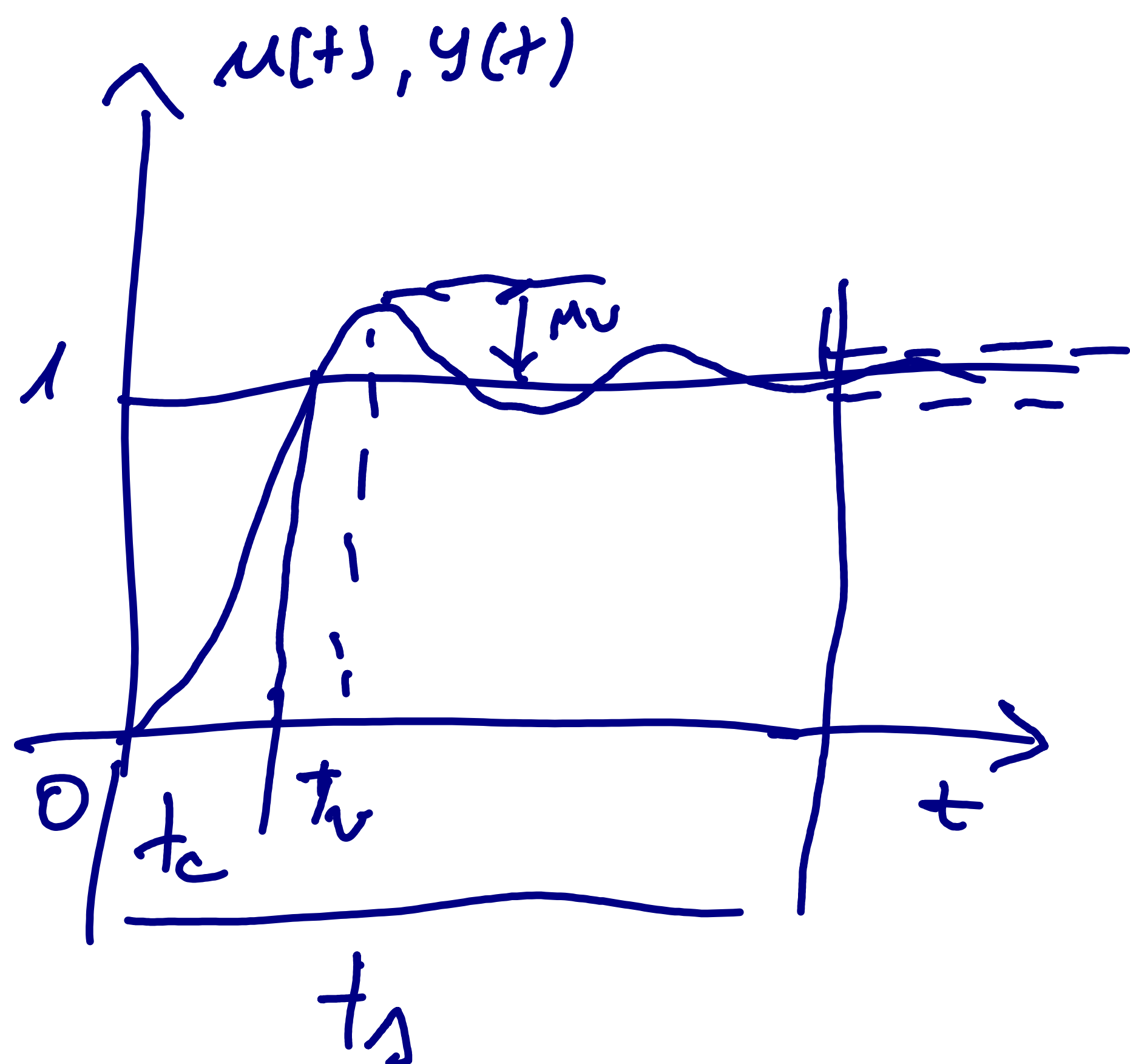
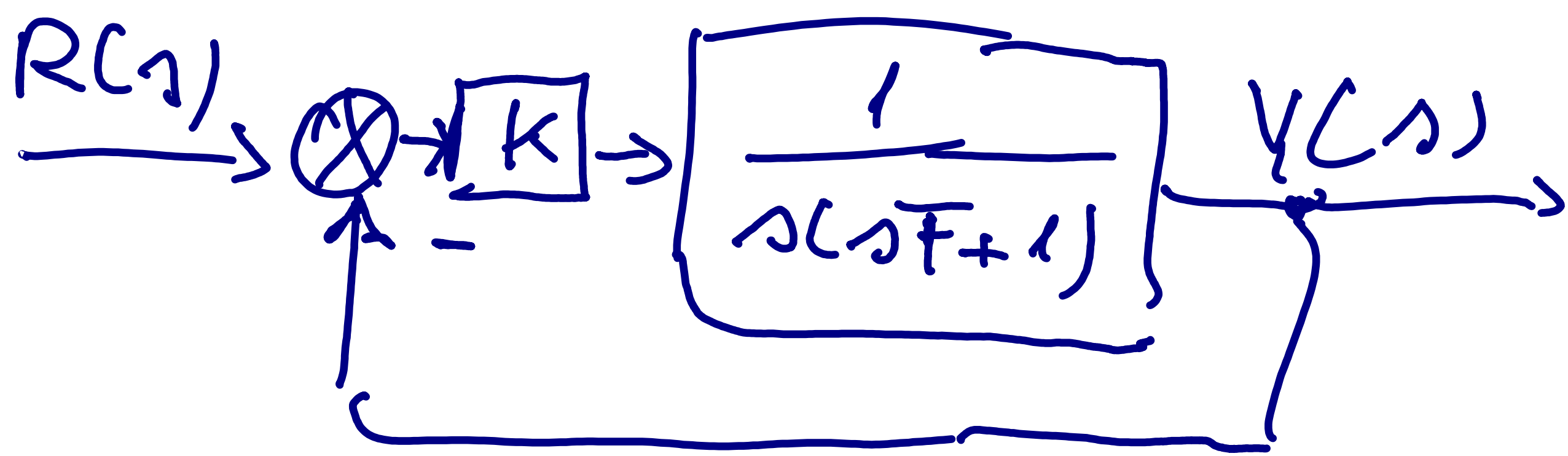
$$k_p = \lim_{s \rightarrow 0} G(s)H(s) = \frac{3}{1} = 3$$

$$t_c \approx 2.2T \Rightarrow 0.5$$

$$t_d \approx 4T (2\%)$$

$$3T (5\%)$$





$$K, T = ?$$

$$t_v = 3s, M_v = 0,25$$

$$\zeta \in (0,1)$$

$$G_o(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$t_c = \frac{\pi - \varphi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$t_v = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$M_v = e^{\frac{-\pi\zeta}{\sqrt{1 - \zeta^2}}}$$

$$t_1 = \frac{4}{\zeta\omega_n} \quad (2\%)$$

$$= \frac{3}{\zeta\omega_n} \quad (5\%)$$

$$\omega_n^2 = \frac{K}{T}$$

$$2\zeta\omega_n = \frac{1}{T}$$

$$M_v \Rightarrow \zeta$$

$$t_v \Rightarrow \omega_n$$

$$G_o(s) = \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s)} = \frac{K}{s^2 T + 1 + K} = \frac{\frac{K}{T}}{s^2 + \frac{1}{T} + K}$$

$$\zeta = \frac{(\ln M_v)}{\sqrt{\pi^2 + \ln^2 M_v}} = 0,4$$

$$\omega_n = \frac{\pi}{t_v \sqrt{1 - \zeta^2}} = 1,14 \text{ rad/s}$$

$$\varphi = \arctg \frac{\sqrt{1 - \zeta^2}}{\zeta}$$

$$t_c < t_v < t_1$$

$$y(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \varphi)$$

Eroarea Stationară

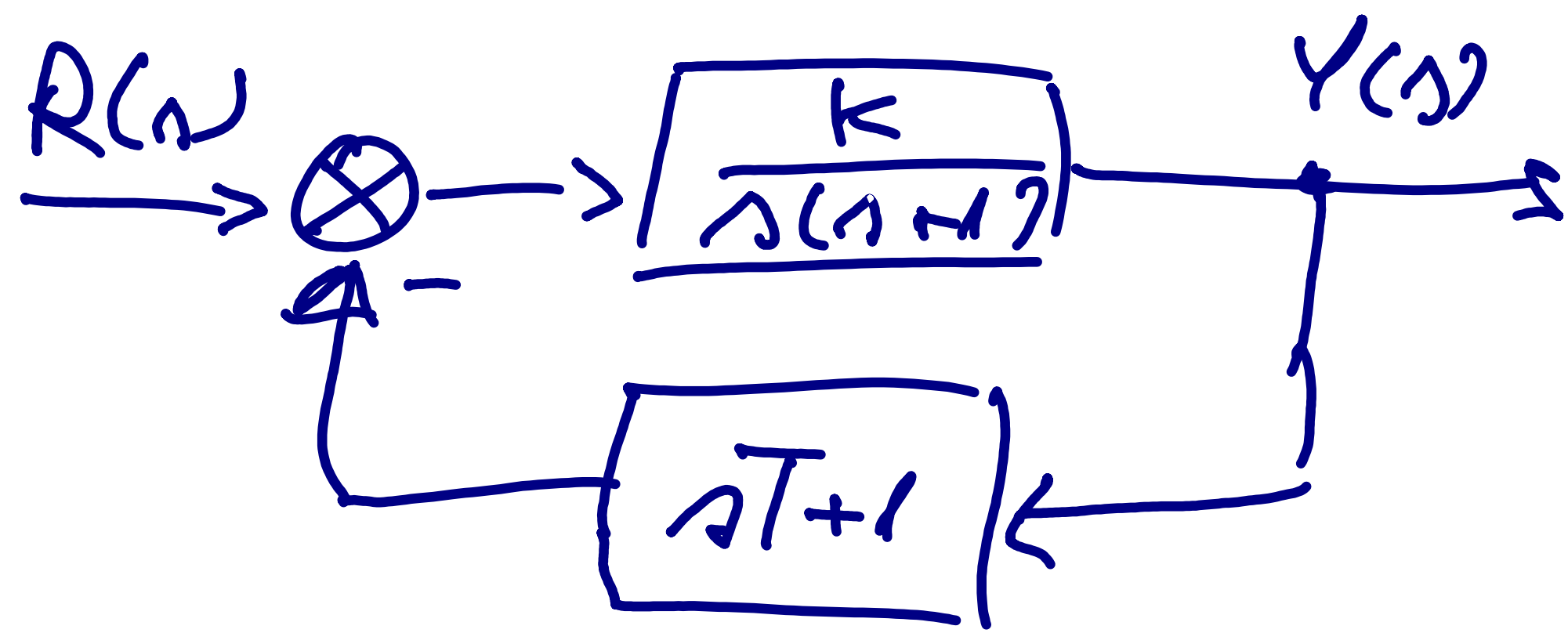
$$G(s)H(s) = G_1(s)G_2(s) = \frac{k}{s^2T + s}$$

$$T = 1,09 \text{ s}$$

$$k = 1,42$$

$$e_{st}^1 = \frac{1}{1+k_p} = \frac{1}{1+\lim_{s \rightarrow 0} \frac{k}{s^2T+s}} = 0$$

$$e_{st}^t = \frac{1}{k_v} = \frac{1}{\lim_{s \rightarrow 0} s \frac{k}{s^2T+s}} = \frac{1}{\lim_{s \rightarrow 0} \frac{k}{sT+1}} = \frac{1}{k}$$



$$M_v = 0,043 \Rightarrow \zeta \in (0,1)$$

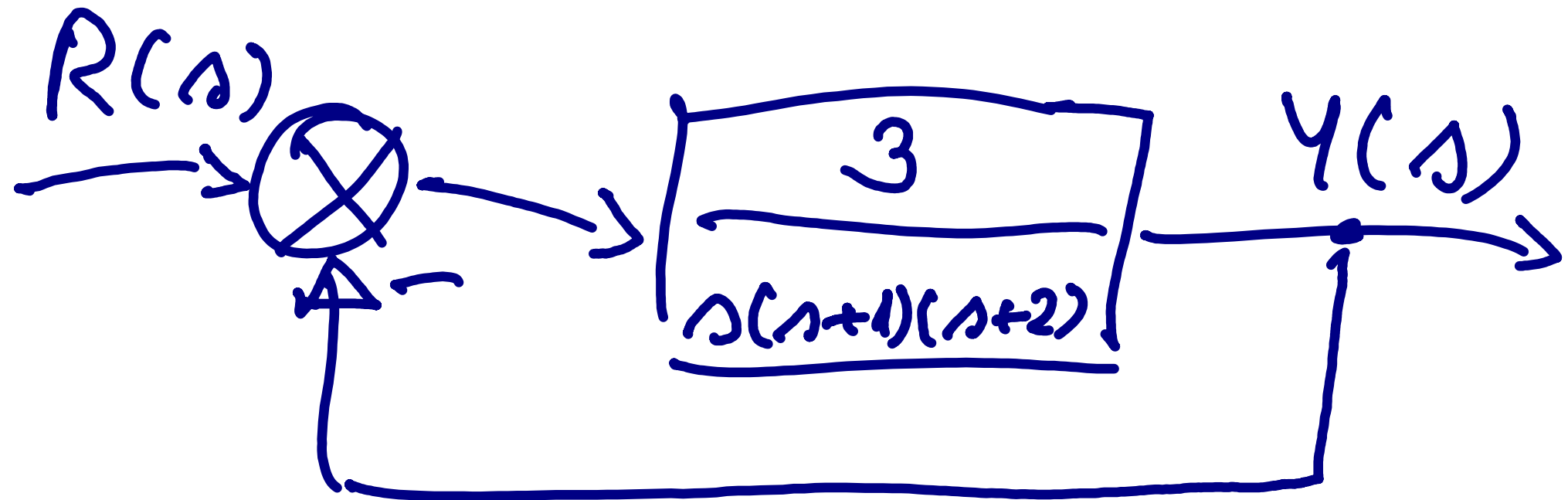
$$t_v = 1 \text{ s}$$

$$G_0(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{k}{s(s+1)}}{1 + \frac{k(sT+1)}{s(s+1)}} = \frac{k}{s^2 + s(1+TK) + k}$$

$$G(s)H(s) = \frac{k(sT+1)}{s(s+1)}$$

$$\Downarrow k = \omega_n^2$$

$$2\zeta\omega_n = 1 + kT$$



$$A(s) = s^3 + 3s^2 + 2s + 3$$

$$k = ?, k > 0$$

$$G_o(s) = \frac{G(s)}{1 + G(s)} = \frac{\frac{3}{s(s+1)(s+2)}}{1 + \frac{3}{s(s+1)(s+2)}} = \frac{3}{s(s+1)(s+2) \cdot \frac{s(s+1)(s+2) + 3}{s(s+1)(s+2)}} = \frac{3}{3 + s(s+1)(s+2)}$$

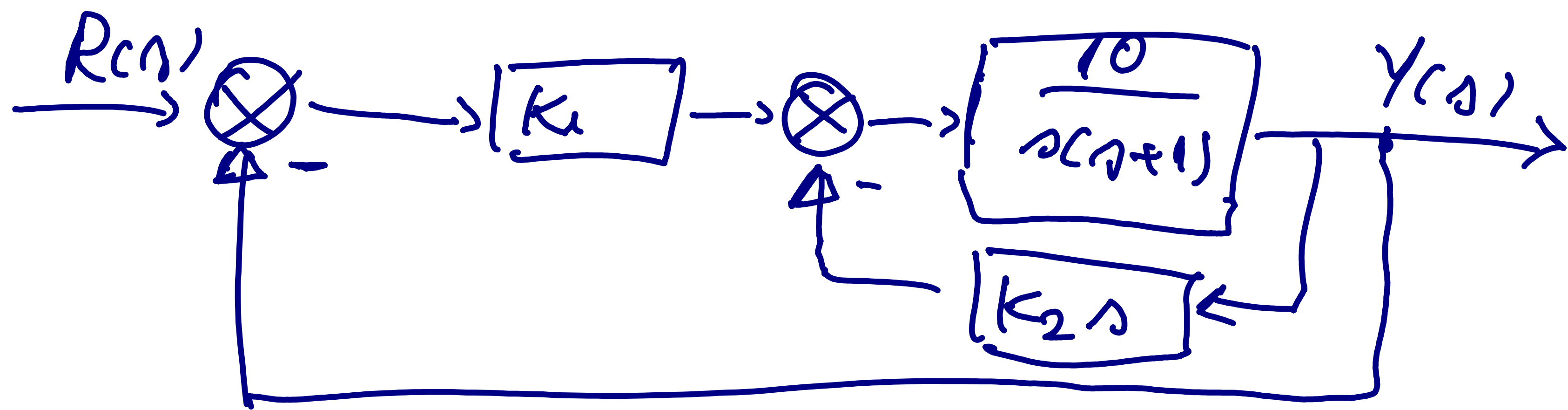
$$= \frac{3}{3 + s(s^2 + s + 2)} = \frac{3}{(s^2 + s)(s+2) + 3} = \frac{3}{s^3 + 3s^2 + 2s + 3}$$

s^3	1	2	0
s^2	3 :1	2 :1	0
s^1	1	0	0
s^0	1	0	0

$$A(s) = s^3 + 3s^2 + 2s + k$$

s^3	1	2
s^2	3	k
s^1	$\frac{6-k}{3}$	0
s^0	k	0

$$\left. \begin{array}{l} \frac{6-k}{3} > 0 \\ k > 0 \end{array} \right| \Rightarrow \begin{array}{l} k < 6 \\ k > 0 \end{array} \Rightarrow k \in (0, 6)$$

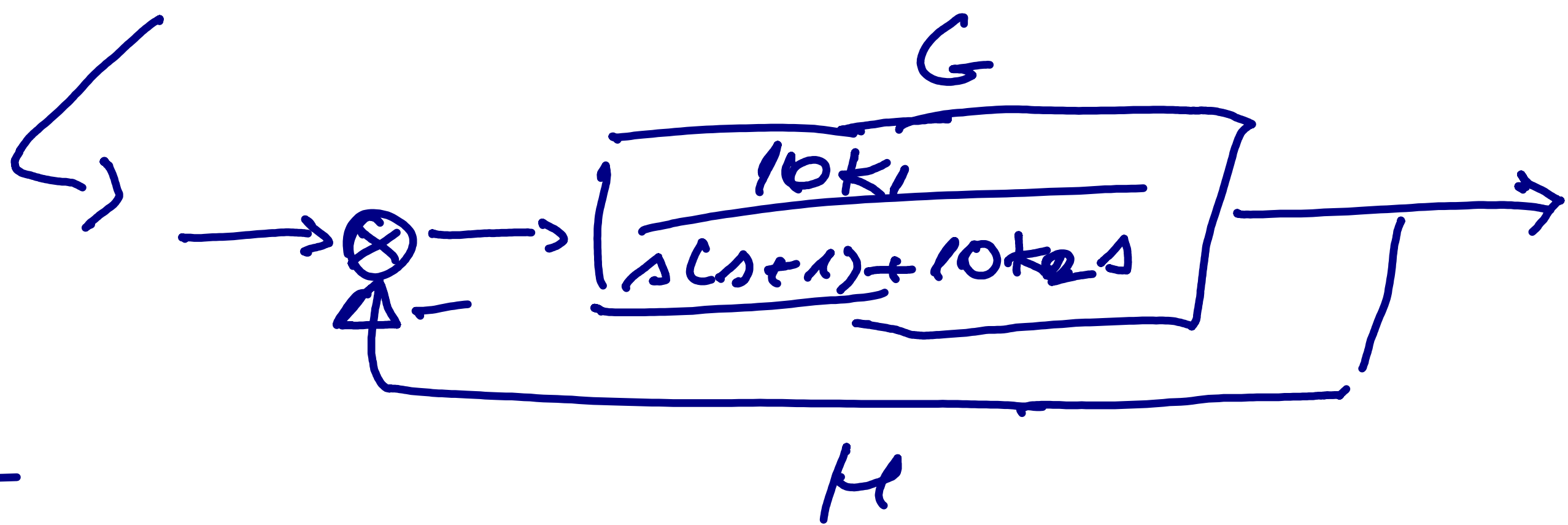
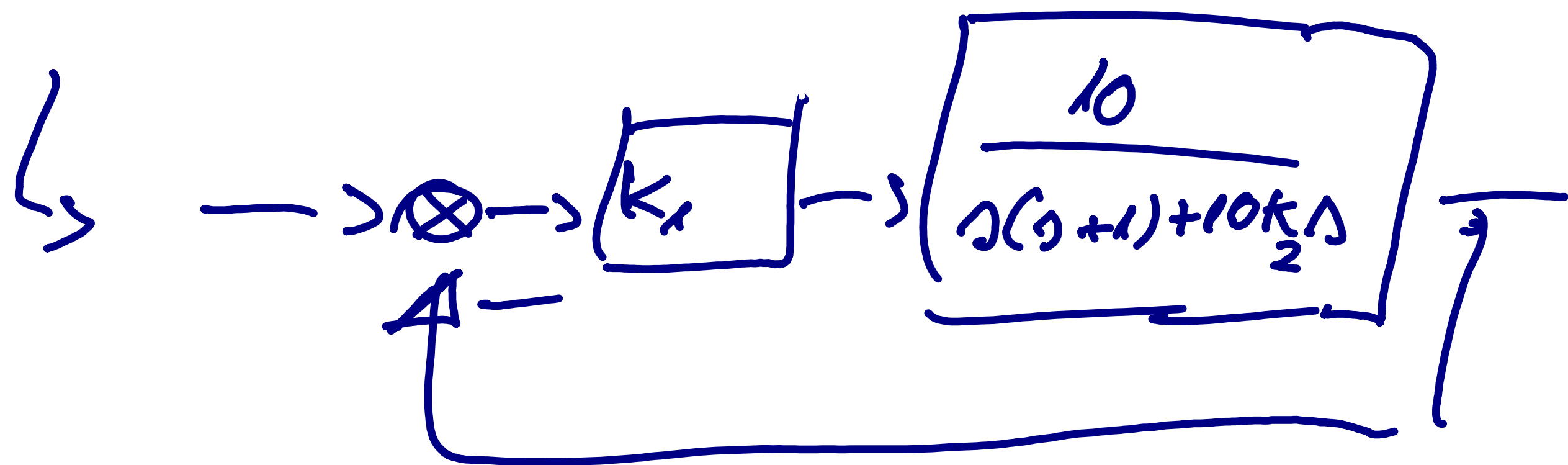


$$K_V = 10, \quad \zeta = 0,5$$

$$K_1, K_2 = ?$$

$$K_V = \lim_{s \rightarrow 0} G(s)H(s)$$

$$G_0 = \frac{G(s)}{1+G(s)} = \underline{\hspace{2cm}}$$



$$G(s) = \frac{10K_1}{s^2 + s(1+10K_2)}$$

$$K_V = \lim_{s \rightarrow 0} sG(s) = \frac{10K_1}{1+10K_2} = 10$$

$$\begin{cases} 2\zeta\omega_n = 1+10K_2 \\ \omega_n^2 = 10K_1 \\ 10K_1 = 10+100K_2 \end{cases}$$